

GROUP ASSIGNMENT 1 – FULL CODE AND DATA AVAILABLE AT
GITHUB.COM/MATHEUSBISSACOT/BIOINFORMATICS

Phylogenetics of a Taste 2 Receptor Member Gene

Gonalo Brochado (20210)^{1, 2} and Matheus Bissacot (202106708)^{1, 2}

¹Faculdade de Engenharia, Universidade do Porto, Street, Postcode, Porto, Portugal and ²Faculdade de Cincias, Universidade do Porto, Street, Postcode, Porto, Portugal

Abstract

Phylogenetic analysis plays a central role in understanding evolutionary relationships between species at the molecular level, functional conservation, and the molecular basis of phenotypic traits [3]. In response to the increasing need for accessible, reproducible, and efficient computational tools, this project proposes a fully automated phylogenetic analysis pipeline developed in Python. Leveraging the Biopython library [2] — a widely adopted suite of tools for computational molecular biology — the system accepts a single protein or nucleotide sequence as input and sequentially performs essential bioinformatic tasks, including sequence preprocessing, BLASTP similarity search [1], multiple sequence alignment, and phylogenetic tree construction. This pipeline provides a valuable resource for researchers seeking to perform robust phylogenetic analyses with minimal manual intervention, addressing both current challenges and best practices in bioinformatics research.

Key words: Phylogenetic, Bioinformatics, Biopython, BLASTP, T2R38

Introduction

Bioinformatics has seen phenomenal growth in the last several years, fueled by concerted implementation of computational methods and informatics tools. Phylogenetics, the study of evolutionary relationships, enables scientists to investigate the divergence of species according to genetic data. Such studies traditionally involved a mix of manual curation and bespoke software, and workflows tended to be fragmented. With the amount of biological data available in public databases still increasing, the need for automatic and easy-to-use pipelines enabling the retrieval of evolutionary information from raw sequences is growing.

The objective of this project was to design and develop a successful pipeline in Python capable of carrying out an entire phylogenetic analysis from a single protein or nucleotide input. Our software handles sequence translation, BLASTP similarity search, multiple sequence alignment, and tree construction. Moreover, the pipeline ensures the generation of standardized and reproducible output files, thereby facilitating downstream analyses and promoting the robustness and reliability of phylogenetic inferences. Particular attention was given to user interaction, computational efficiency, and the modularity of the system, allowing for future extensions and adaptability to a wide range of biological datasets.

Implementation

The implementation of the proposed pipeline was conducted with a strong emphasis on modularity. The pipeline was

structured into distinct stages, each corresponding to a key step in the phylogenetic analysis: preprocessing and translation of nucleotide sequences, homology search through BLASTP, multiple sequence alignment (MSA), and phylogenetic tree reconstruction.

Tools and Packages

To implement the phylogenetic analysis pipeline, a range of bioinformatics tools and Python packages were employed. These tools were selected based on their popularity, reliability, and seamless integration with Python.

- **Biopython:** The core bioinformatics library used for sequence manipulation, BLAST querying and parsing, multiple sequence alignment handling, and phylogenetic tree construction. Key modules used include `Bio.Seq`, `Bio.Blast`, `Bio.AlignIO`, and `Bio.Phylo`.
- **NCBI BLASTP:** Used to perform protein similarity searches against the NCBI non-redundant (nr) database. It was accessed programmatically through the `Bio.Blast.NCBIWWW` interface, returning results in XML format.
- **EBI Clustal Omega API:** Multiple sequence alignments were conducted through the European Bioinformatics Institute's RESTful Clustal Omega service. This cloud-based solution avoided the need for local software installation.
- **DistanceTreeConstructor (Biopython):** Used to construct a phylogenetic tree from aligned sequences using the

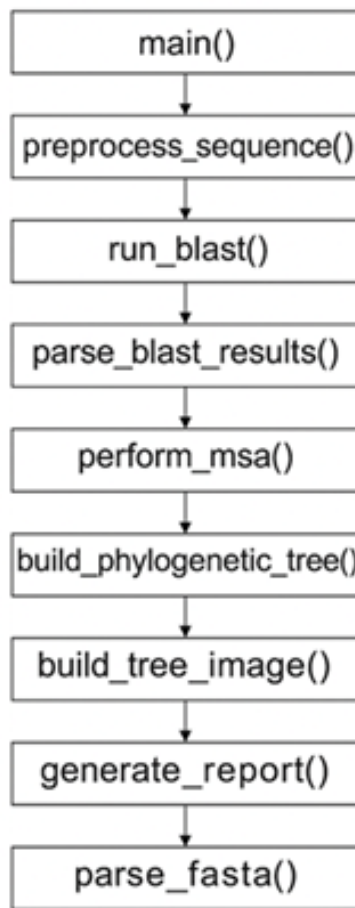


Fig. 1. Pipeline

Neighbor-Joining (NJ) algorithm, leveraging BLOSUM62-based distance metrics.

- **Matplotlib:** Employed to render and save the phylogenetic tree image in a visual format (.png).
- **Jinja2:** Template engine used to dynamically generate an HTML report based on the collected data from each pipeline stage.
- **xhtml2pdf (pisa):** Converts the rendered HTML report into a downloadable PDF file for documentation and sharing purposes.
- **Standard Python Libraries:** Modules such as `argparse`, `os`, `uuid`, `requests`, and `datetime` were used for command-line argument parsing, HTTP requests, unique file naming, filesystem operations, and timestamping.

Challenges

The development of the phylogenetic pipeline presented several key challenges that required careful consideration. A primary obstacle involved ORF detection and sequence ambiguity, as many DNA inputs contained incomplete codons or ambiguous bases (particularly 'N' nucleotides). To address this, the `preprocess_sequence()` function implemented six-frame translation coupled with user-guided ORF selection, ensuring biological accuracy at the cost of full automation.

Processing BLASTP outputs introduced another layer of complexity due to inconsistent species naming conventions,

which was resolved through advanced pattern matching and multi-level filtering in `parse_blast_results()` to maintain phylogenetic diversity. The asynchronous nature of the EBI Clustal Omega REST API necessitated a robust polling system with network error recovery in `poll_clustal_status()` to handle variable processing times.

Initial tree construction attempts revealed compatibility issues with Clustal-generated .dnd files, leading to a switch to custom distance matrices using BLOSUM62 and Biopython's `DistanceTreeConstructor` for improved reliability.

While the pipeline achieves near-complete automation, the deliberate retention of manual ORF verification ensures biological validity, though future implementations could incorporate domain conservation scoring or machine learning approaches for full automation.

These solutions collectively yielded a robust analytical tool that balances automation with necessary biological verification steps.

Final Code

The phylogenetic analysis pipeline was implemented as a modular Python program, composed of two main files: `phylo-analysis.py` and `utils.py`. The former serves as the execution entry point, while the latter contains utility functions responsible for data processing and analysis. Below is a breakdown of the key functions and their responsibilities:

Main Components

- **main():** Orchestrates the entire pipeline by calling each step in sequence
- **preprocess_sequence():** Handles input sequence reading and ORF detection
- **run_blast():** Performs BLASTP searches against NCBI's nr database
- **parse_blast_results():** Processes BLAST outputs and selects diverse homologs
- **perform_msa():** Conducts multiple sequence alignment via EBI Clustal Omega
- **build_phylogenetic_tree():** Constructs phylogenetic trees from alignments
- **build_tree_image():** Renders the phylogenetic tree from XML format
- **generate_report():** Creates analysis summary and output verification
- **parse_fasta():** Parses the FASTA file of homologous sequences and extracts species identifiers and sequence descriptions into a structured list.
- **parsealn_file():** Reads the Clustal alignment file, reconstructs complete sequences per species, and ensures consistent visual alignment formatting.
- **create_report():** Loads and renders an HTML report template using Jinja2, then exports the results to HTML.

Each of these functions is modular, reusable, and designed to be easy to debug or expand. Together, they enable automated retrieval, alignment, and comparison of homologous proteins, culminating in the generation of a visualizable phylogenetic tree.

main()

Located in `phylo-analysis.py`, this function orchestrates the entire pipeline by calling each step in sequence. It takes two arguments from the command line: the path to the input

FASTA file and the number of species to include in the phylogenetic analysis. It handles the overall logic, user input, and flow control.

preprocess_sequence(input_file)

This function reads the input FASTA file, removes ambiguous nucleotides (like 'N'), and identifies the longest open reading frame (ORF) by translating the sequence in all six reading frames. It then allows the user to select the preferred frame manually. This provides flexibility and ensures the use of the most biologically relevant peptide sequence.

run_blast(protein_sequence)

Executes a BLASTP search using NCBI's online tools, submitting the protein sequence and retrieving the top results. The XML output is stored locally for later parsing. This step identifies homologous sequences from different species for comparative analysis.

parse_blast_results(blast_xml, num_species, human_protein_seq)

Parses the BLAST XML output to extract protein sequences from a diverse set of species. The number of unique species to include is user-defined. It writes these sequences (along with the human sequence) into a new FASTA file to be used for multiple sequence alignment (MSA).

perform_msa(input_fasta)

Performs a MSA using the EBI Clustal Omega web service. The function handles submission, polling for job completion, and fetching of alignment results in Clustal format. This alignment is critical for evaluating evolutionary relationships.

build_phylogenetic_tree(msa_file)

Constructs a phylogenetic tree using the Neighbor-Joining (NJ) algorithm. It reads the aligned sequences, calculates a distance matrix using the BLOSUM62 substitution model, and outputs the tree in PhyloXML format. This tree reflects evolutionary distances between species based on the aligned protein sequences.

build_tree_image(tree_file)

Generates a visual representation of the phylogenetic tree from the XML file. This step is essential for inclusion in the final report and for human interpretation of the relationships among species.

generate_report(...)

Produces a summary .txt file that logs paths to key output files and confirms the successful execution of the pipeline. It acts as a simple trace for validation or troubleshooting.

parse_fasta(fasta_file, human_protein)

Reads the homologous sequences in the FASTA format and extracts identifiers and descriptions. These are stored in a structured list for later display in the final HTML/PDF report.

parse_aln_file(filepath)

Parses the Clustal alignment file to reformat and align sequences consistently across all entries. It helps in generating a clean, visual alignment matrix suitable for display and analysis in the report.

create_report(data)

Uses Jinja2 to load an HTML template (report.html), renders it using the collected data, and then converts it into both an HTML. This final step automates documentation of the analysis and ensures reproducibility and presentability.

Together, these functions implement a full-fledged phylogenetic pipeline capable of taking a raw DNA/protein sequence and producing biologically relevant insights, including evolutionary relationships and alignment data—all in a fully automated, reproducible manner.

Results

The T2R38 gene was selected for molecular phylogenetic investigation. This receptor is implicated in the perception of phenylthiocarbamide (PTC) and other bitter compounds, and exhibits considerable polymorphic variation across populations and species. In this section we will analyze the results we obtained from the analysis of this gene.

Translation Frame

Following rigorous examination of the six potential open reading frames, frame 3 was identified as the most biologically plausible translation as we can see here VLAWNVLHPFCT GWQPGL*ISQLEKRSRIAN*RSIMLTLTRITVSYEVRST FLFISVLEFAVGFLTNAFVFLVNFWDVVKRQALSNSDCV LLCLISIRLFLHGLLFLSAIQLTHFQKLSEPLNHSYQAIIML WMIANQANLWLAACLSLLYCSKLIRFSHTFLICLASWVSR KISQMLLGILCSCICTVLCVWCFFSRPHFTVTTVLFMNN NTRLNWQIKDLNLFYSFLFCYLWSVPFLLFLVSSGMLT VSLGRHMRMTMKVYTRNSRDPSEAHIKALKSLVSFFCFF VISSCAAFISVPLLLWRDKIGVMVCVGIMAACPSGHAAIL ISGNAKLRRAVMTILLWAQSSLKVRADHKADSRITLC*EW T*NELFINTPVSLHXYASDSSGIQL*FLTKPSNFFYKTKLQK VSPLCINEL. The translated peptide sequence from this frame demonstrated structural integrity and absence of premature stop codons over a significant span, indicative of a valid coding region.

BLAST Results

After the decision of the best frame, a BLAST search was carried out against the non-redundant protein database (nr) to identify orthologous proteins. The analysis yielded 30 high-confidence hits, all exhibiting significant sequence homology (E-values $\leq 1e-50$) with known T2R38 orthologs. We decided to search for 3 times the number of different species we wanted, which in this case was 10. The top alignments corresponded to Homo sapiens, Pan troglodytes, Gorilla gorilla, and other primates, reinforcing the notion of evolutionary conservation within the Hominidae clade. The retrieved homologous sequences predominantly belong to higher primates, underscoring the evolutionary constraints acting upon this gene. Minimal sequence divergence was observed among humans, chimpanzees, and gorillas, consistent with their recent common ancestry.

Multiple Sequence Alignment

A multiple sequence alignment (MSA) was performed using Clustal Omega to assess sequence conservation and divergence patterns across taxa. The alignment revealed extensive conservation in the seven-transmembrane domain regions, a hallmark of class A GPCRs. Variability was more pronounced

in the N- and C-terminal regions, which are typically less constrained and more permissive to mutational events. The overall pattern supports strong purifying selection acting on functional domains, with potential evolutionary plasticity in peripheral regions.

Phylogenetic Tree

The phylogenetic analysis of the T2R38 orthologs and their related sequences showed a structure that aligns with established primate evolutionary connections as we can see in Figure 2. The tree lacks roots and consists of three main clades. The initial significant division categorizes the Homo sapiens input sequence alongside a synthetic construct and a reference Homo sapiens sequence (seq2), creating a strongly supported subclade with minimal branch lengths, reflecting high sequence similarity. This cluster is, in turn, related to Gorilla gorilla (seq4), showing slight divergence (branch length 0.0082), indicating an evolutionary closeness within the hominid lineage.

A second clearly defined clade groups Pan troglodytes (seq3) and Pan paniscus (seq5), showing little divergence between them (branch length 0.005), consistent with predictions from great ape phylogeny. Along with the hominid sequences, these taxa constitute a unified homininae superclade, reinforcing the functional conservation of the T2R38 receptor among these species.

The third primary lineage includes Pongo pygmaeus (seq7 and seq10) and the smaller apes, such as Hylobates klossii (seq9), Symphalangus syndactylus (seq8), and Nomascus leucogenys (seq6). This group shows increased internal variation (e.g., branch lengths reaching approximately 0.0172 for Hylobates), indicating more profound evolutionary separations within the hominoid lineage. Significantly, the separation between Pongo sequences and the lesser apes aligns with fossil findings that suggest the orangutan divergence occurred earlier in primate evolution.

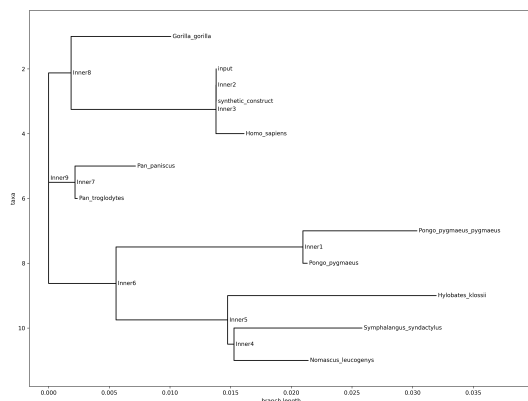


Fig. 2. Phylogenetic Tree

Report in HTML

To make it easier for the end user to analyze this script, an HTML is created with all the results obtained from the analysis.

Conclusion

Although the analysis is intricate and the execution time of the pipeline is lengthy, the integrative pipeline applied in this study effectively facilitated the identification, translation, comparative analysis, and phylogenetic contextualization of the T2R38 gene across multiple primate species. The choice of the most biologically viable open reading frame confirmed the legitimacy of the inferred protein product, which was then utilized in a strong alignment of multiple sequences and within a phylogenetic inference framework. The resulting evolutionary framework demonstrated strong agreement with recognized primate lineages and validated the preservation of the T2R38 gene among hominids, highlighting a notable similarity between the Homo sapiens and Pan species. Furthermore, the addition of synthetic and non-human primate sequences offered an extensive perspective on sequence divergence and functional preservation. Together, these findings highlight the efficacy of computational pipelines in clarifying molecular evolution and confirm the phylogenetic significance of taste receptor genes in primate classification.

The script can also be aimed at the end user, allowing them to conveniently access a report of the analysis without needing to examine the outcomes of the various steps individually.

References

1. Stephen F Altschul, Thomas L Madden, Alejandro A Schaffer, Jinghui Zhang, Zheng Zhang, Webb Miller, and David J Lipman. Gapped blast and psi-blast: a new generation of protein database search programs. *Nucleic acids research*, 25(17):3389–3402, 1997.
2. Peter JA Cock, Tiago Antao, Jeffrey T Chang, Brad A Chapman, Cymon J Cox, Andrew Dalke, Iddo Friedberg, Thomas Hamelryck, Konrad Kauff, Bartek Wilczynski, and Michiel J L de Hoon. Biopython: freely available python tools for computational molecular biology and bioinformatics. *Bioinformatics*, 25(11):1422–1423, 2009.
3. Ziheng Yang and Bruce Rannala. Molecular phylogenetics: principles and practice. *Nature Reviews Genetics*, 13(5):303–314, 2012.